MAGeCKrs provides deterministic and auditable Rust workflows for CRISPR screen analysis

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Manuscript and Supplementary Information

Scientific figures included

1 Abstract

MAGeCKrs implements guide counting, robust rank aggregation (RRA), and maximum likelihood estimation (MLE) in Rust, providing deterministic CPU execution and explicit fit diagnostics. We evaluated three workflows from the E-MTAB-13979 Yusa v3 screen: PC-9 RRA, PC-9 MLE (Day 10 contrast), and HCC827 MLE (Day 23 contrast). Raw guide counts agreed exactly with Legacy MAGeCK across all evaluated settings. In PC-9 RRA, median complete-workflow runtimes were 623.1 seconds for Legacy MAGeCK and 165.1 seconds for MAGeCKrs (3.77-fold median paired ratio across three fresh pairs), with substantial but incomplete ranking overlap; additional MAGeCKrs rows reflect output-retention rules rather than discoveries. In a separate campaign (six runs per engine), PC-9 MLE median runtimes were 242.0 and 65.9 seconds (3.67-fold ratio of medians) with close beta agreement (Spearman correlation 0.999996). For HCC827 MLE, median runtimes were 238.0 and 65.1 seconds (3.65-fold ratio of medians), showing high beta-rank and sign agreement across 18,542 common genes (Spearman correlation 0.997660; 18,534 matching signs) alongside selected large coefficient and Wald-inference differences. These results demonstrate faster workflow completion and auditable execution in MAGeCKrs while defining clear boundaries on cross-implementation equivalence.

2 Introduction

Pooled CRISPR knockout screens link genetic perturbations to phenotypes by measuring changes in guide-RNA abundance across conditions. Identifying reliable gene-level effects requires separating consistent multi-guide signals from guide-specific variation, sequencing noise, and library representation. MAGeCK addresses these needs by combining count normalization and negative-binomial guide modeling with modified robust rank aggregation (RRA) [1, 3], while MAGeCK-VISPR extends the framework with maximum likelihood estimation (MLE) for multi-condition designs [2, 6].

Reimplementing established screening workflows makes reproducibility an empirical question: FASTQ grouping defines the sample matrix, guide filtering determines the analyzed universe, and numerical stopping rules govern whether iterative fits yield valid inference. Because runtime gains are meaningful only alongside the outputs they produce, speed and analytical agreement must be evaluated together. Here we present MAGeCKrs, a Rust implementation of core MAGeCK workflows: COUNT, RRA through the `test` command, and default scalar MLE. Using an experimental osimertinib screen in human lung cancer cell lines [5], we evaluate complete counting-to-analysis workflows against Legacy MAGeCK across execution speed, numerical repeatability, and scientific agreement, without asserting new biological hit sets.

3 Design and implementation

MAGeCKrs separates input validation, count-matrix construction, shared statistical routines, gene-level analysis, and output writing (Figure 1). COUNT maps grouped FASTQ reads to a guide library, generating count matrices and physical-read quality summaries. RRA/`test` computes directional guide statistics between control and treatment samples and aggregates them to genes. MLE aligns count tables with a design matrix and reports condition coefficients with ordinary Wald inference.

Optional worker threads parallelize independent MLE gene fits while preserving the scalar model, and RRA scheduling maintains a fixed random-number and reduction sequence. Results and diagnostic errors are assembled in input source order. These choices ensure repeatable execution under the tested settings without guaranteeing identical outputs across arbitrary hardware or future inputs.

In MLE, MAGeCKrs requires finite, converged coefficients and valid Wald statistics at the returned vector. When a primary fit does not converge, reduced-relaxation retries re-estimate the same model from the original initialization without altering dispersion or ridge parameters. Unresolved planned

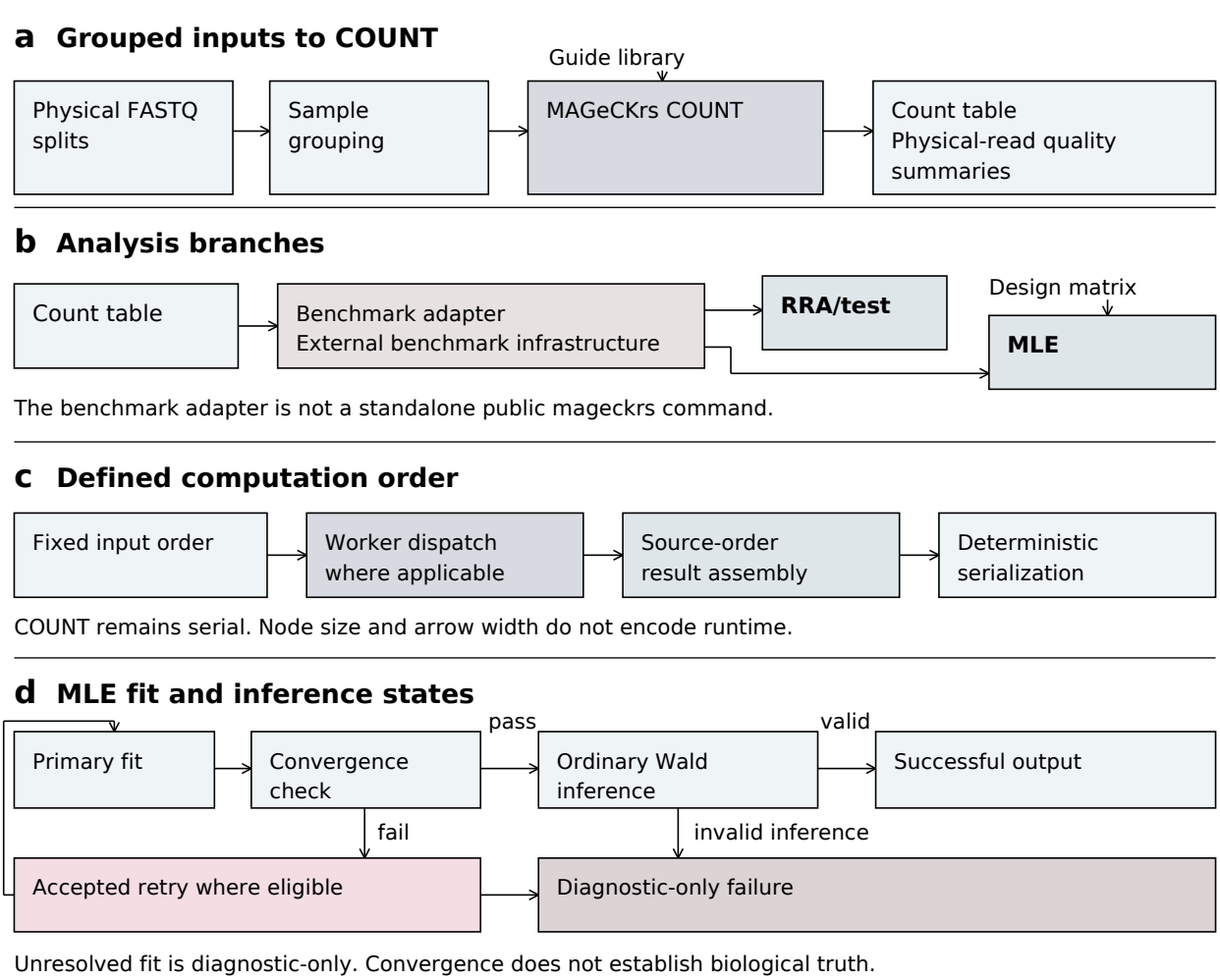


Figure 1. MAGeCKrs workflow and architecture. (a) COUNT processes grouped FASTQs and a guide library into count tables and quality summaries. (b) In the evaluated workflows, a canonical adapter prepares counts for RRA/`test` or design-matrix MLE. (c) Defined input ordering and source-order assembly provide repeatable outputs under the tested configurations. (d) Convergence, ordinary Wald inference, and diagnostic-only failure are distinct states; unresolved fits trigger relaxation retries or exit as diagnostic-only failures without unverified summaries. Box and arrow dimensions are schematic and do not encode runtime. Details are in Supplementary Methods.

hypotheses emit fit diagnostics without writing unverified summaries. Output metadata record fit attempts, excluded genes, inference populations, and the effective RRA seed. Permutation-style MLE columns are fixed placeholders and must not be interpreted as computed inference.

Additional interfaces for exact UMI counting, COUNT-to-`test` orchestration, and a copy-number variation (CNV) adapter fall outside the benchmarks reported here (Supplementary Table 5).

4 Benchmark design

We evaluated Legacy MAGeCK and MAGeCKrs on three workflows from the Yusa v3 subset of E-MTAB-13979, an osimertinib screen in lung cancer cell lines [5] using the Ong et al. guide library [4, 7]. PC-9 RRA and PC-9 MLE compare PC-9 Day 10 osimertinib (7.5 nM) against DMSO; HCC827 MLE compares HCC827 Day 23 osimertinib (10 nM) against DMSO. Each contrast uses two replicates (R1/R2) per condition (Table 1).

Workflows differ in their counted inputs and statistical models. PC-9 RRA counts 20 compressed FASTQ files grouped into ten samples before selecting the four Day 10 contrast samples for RRA. PC-9 MLE counts eight FASTQ files grouped into the four PC-9 Day 10 samples, and HCC827 MLE counts eight FASTQ files grouped into the four HCC827 Day 23 samples. Thus, PC-9 RRA and PC-9 MLE evaluate the same biological contrast from different counted inputs. Physical FASTQ splits are summed within sample groups, and both MLE designs include an intercept and a treatment coefficient.

PC-9 RRA comprises three fresh paired rounds without warmup or run replacement, using the median within-pair runtime ratio as primary estimator. PC-9 MLE and HCC827 MLE each comprise six valid observations per engine from a separate campaign, evaluated via ratios of engine medians. Campaign observations are not pooled; provenance appears in Methods and the supplement.

5 Results

5.1 Exact counts precede nonidentical gene outputs

Raw count matrices were identical between engines across all three workflows, providing a common input foundation for evaluating downstream differences. Across 114,653 canonical guides, validation found zero discrepant cells: 1,146,530 cells in PC-9 RRA and 458,612 cells in each of PC-9 MLE and HCC827 MLE. A shared canonical adapter retained 113,806 guides (112,802 targeting guides and 1,004 controls), excluding 847 guides and omitting 200 fully lost targets without zero-filling them. Each run used its freshly counted table, matching downstream reference matrices byte for byte (Figure 3).

Despite identical counts, native PC-9 RRA table sizes differed: Legacy MAGeCK emitted 18,187 gene rows, whereas MAGeCKrs emitted 18,543. The 356 additional MAGeCKrs rows comprise one Control row and 355 all-zero-in-contrast genes (distinct from the 200 lost targets). These rows reflect output retention rather than biological discoveries; comparisons use the 18,187-gene common universe.

5.2 PC-9 RRA performance and output comparison

MAGeCKrs completed the PC-9 RRA workflow faster than Legacy MAGeCK across all three fresh pairs. Complete-workflow median times were approximately 623.1 seconds for Legacy MAGeCK and 165.1 seconds for MAGeCKrs, yielding a 3.77-fold median paired ratio through final output closure (Figure 2; exact values in Table 2). Serial COUNT dominated runtime; stage breakdowns appear in Supplementary Table 3 and Supplementary Figure 1.

Within the 18,187-gene common universe, PC-9 RRA rankings showed substantial but incomplete overlap: 82/100 and 410/500 in the negative direction, and 80/100 and 397/500 in the positive direction (identical under native and common rankings). Continuous scores, p-values, FDRs, ranks, and fold changes differed. Maximum absolute FDR differences reached 0.945401 and 0.789528212 for the negative and positive directions, demonstrating that rank overlap does not imply significance equivalence.

5.3 PC-9 MLE performance and coefficient agreement

In PC-9 MLE, MAGeCKrs reduced runtime while producing treatment coefficients that closely matched Legacy MAGeCK. Across six observations per engine, median runtimes were approximately 242.0 and 65.9 seconds (a 3.67-fold ratio of medians). Treatment betas showed a Spearman correlation of 0.999996 and matching signs across all 18,542 genes, with a maximum absolute beta difference of 0.04862 for U2AF2. This close agreement does not imply exact equivalence (Table 3).

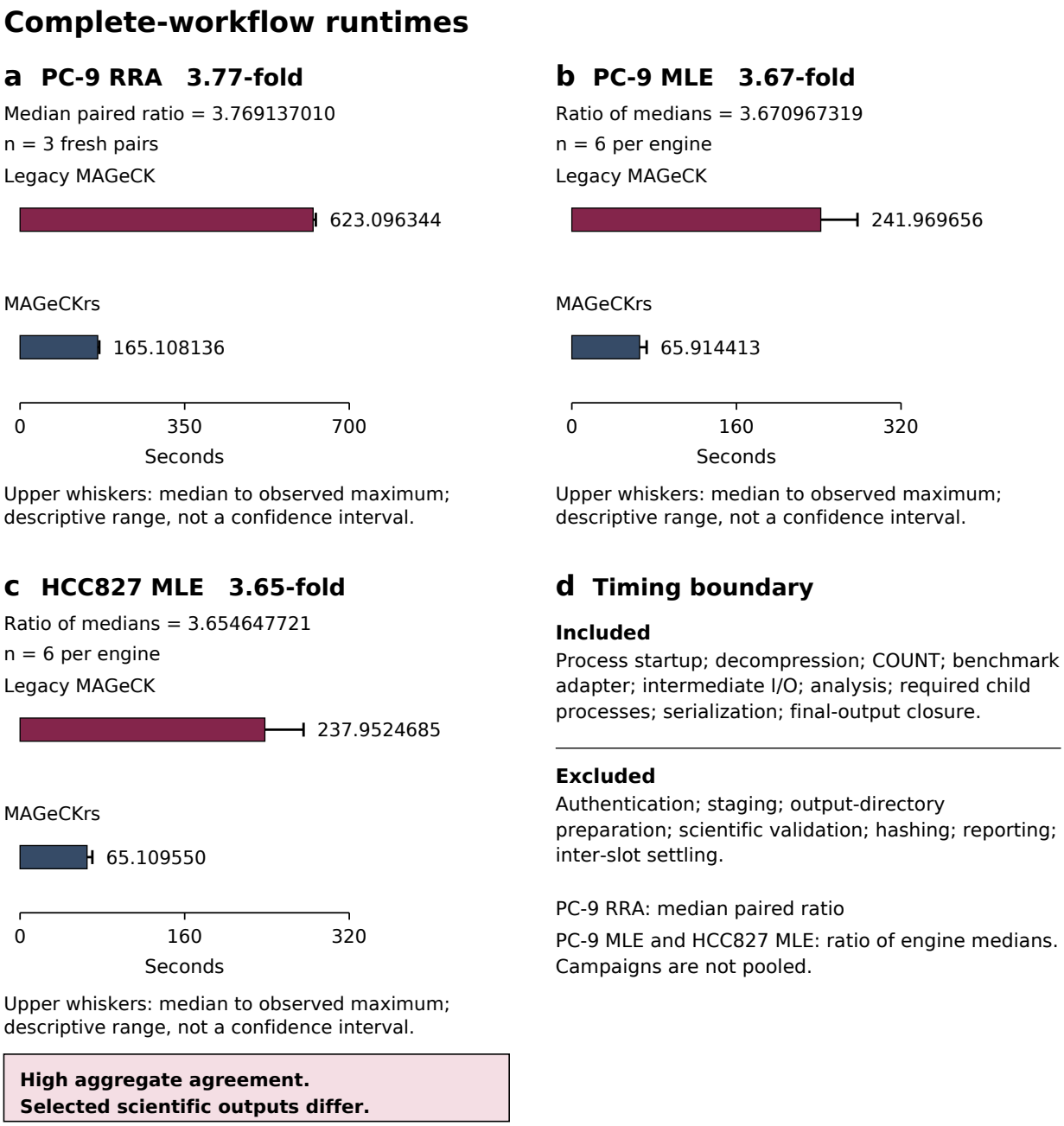


Figure 2. Complete-workflow runtimes. (a) PC-9 RRA across three fresh paired rounds (median paired ratio). (b) PC-9 MLE across six runs per engine (ratio of medians). (c) HCC827 MLE across six runs per engine (ratio of medians; differing scientific outputs indicate workflow completion rather than an identical-output speedup). (d) Operations within the timing boundary and primary ratio estimators. Solid bars show median runtimes; black one-sided whiskers extend to observed maxima without lower whiskers. Whiskers indicate descriptive observed ranges, not confidence intervals; individual runs appear in Supplementary Table 2 rather than overlaid. Campaigns are not pooled. Complete minima and maxima appear in Table 2 and Supplementary Table 2; HCC827 MLE output differences are in Figure 4 and Supplementary Table 1, and PC-9 RRA stages appear in Supplementary Figure 1.

5.4 HCC827 MLE agreement and selected material differences

In HCC827 MLE, median complete-workflow runtimes were approximately 238.0 seconds for Legacy MAGeCK and 65.1 seconds for MAGeCKrs across six observations per engine (a 3.65-fold ratio of medians). Because the two tools produced different estimates on a subset of genes, this ratio measures completion of the fixed workflow rather than identical-output execution.

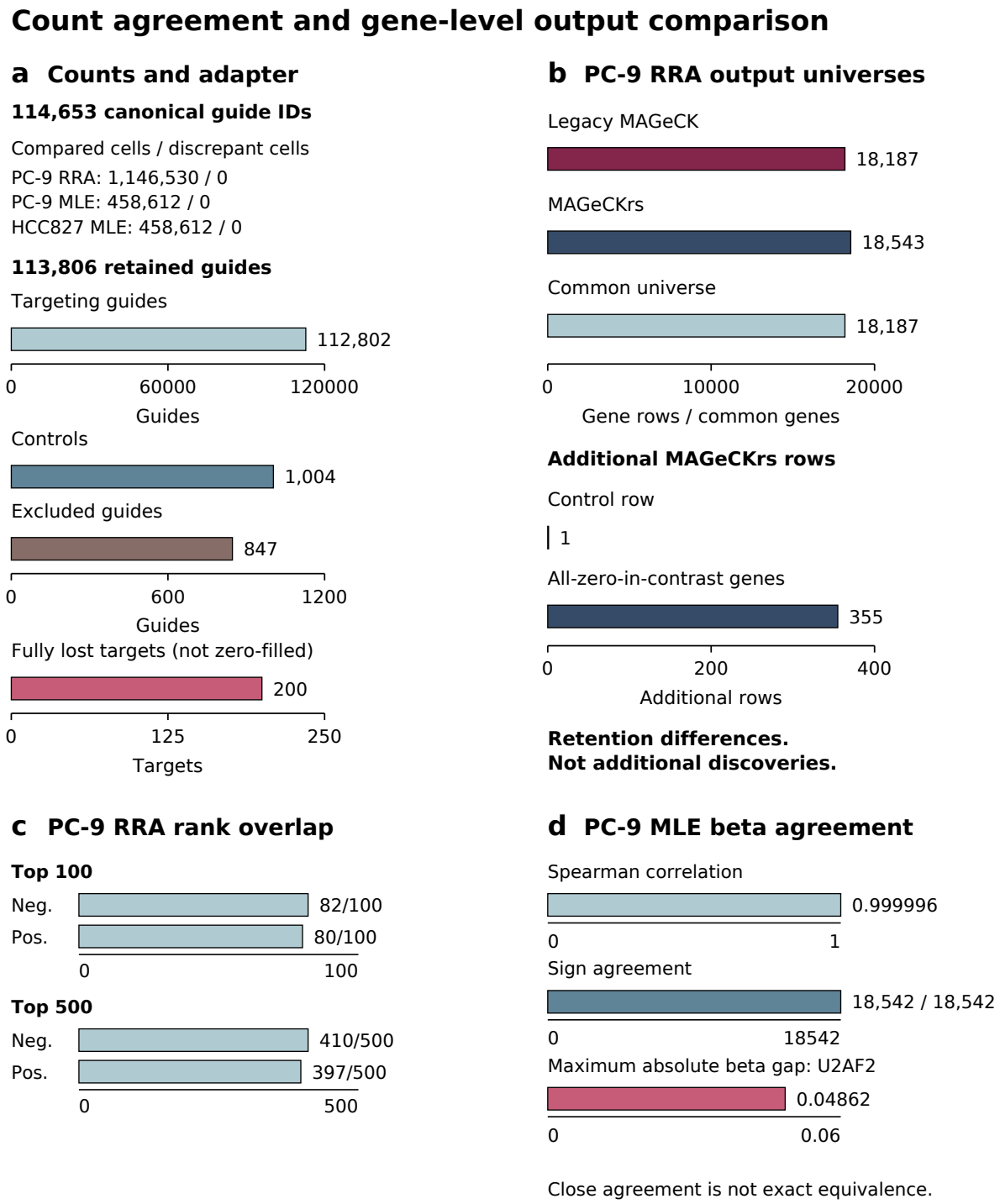


Figure 3. Count agreement and gene-level output comparison. (a) Raw COUNT validation across 114,653 canonical guides yielded zero discrepant cells in all three workflows, followed by canonical adapter filtering to 113,806 retained guides (112,802 targeting and 1,004 controls). (b) In PC-9 RRA, both engines shared 18,187 targeting genes; 356 additional MAGeCKrs rows (one Control row and 355 all-zero contrast genes) reflect output-retention conventions rather than discoveries. (c) PC-9 RRA top-ranked gene overlap reached 82/100 and 410/500 (negative) and 80/100 and 397/500 (positive) under native and common-universe ranking. (d) In PC-9 MLE, treatment beta coefficients showed Spearman correlation 0.999996, matching signs across all 18,542 genes, and a maximum absolute difference of 0.04862 (U2AF2). Horizontal bars use panel-specific scales without error bars or inferred scatter.

Across all 18,542 common genes, Legacy MAGeCK and MAGeCKrs showed high rank and directional agreement, with a beta Spearman correlation of 0.997660 and matching beta signs for 18,534 genes. Linear magnitude and Wald-statistic agreement were lower across the full population: beta Pearson correlation was 0.658132 and beta root mean square error (RMSE) was 0.620440. The z-statistic Pearson correlation was 0.660335 with a maximum absolute z difference of 132.57117. High genome-wide rank and sign consistency therefore coexisted with selected large coefficient and Wald-inference differences (Figure 4).

TOMM20L, TMEM151B, and NMRK2 illustrate these large divergences; they are selected extreme examples and are not representative of all 18,542 common genes. Estimated betas in Legacy MAGeCK versus MAGeCKrs were -58.283 versus 0.46517 for TOMM20L (absolute gap 58.74817). They were -55.254 versus -0.18546 for TMEM151B (gap 55.06854) and -22.119 versus 0.1786 for NMRK2 (gap 22.29760). Legacy MAGeCK printed Wald p/FDR values were 0/0 for all three genes, whereas MAGeCKrs reported p/FDR values of 0.37284/0.59948, 0.60177/0.7627, and 0.61899/0.77497, respectively. Legacy MAGeCK printed these p-values and FDRs as zero; the printed zeros do not imply zero biological uncertainty.

The complete seven-gene selected-difference panel, including MAP2K1, LMO4, CCDC12, and RPL36, appears in Supplementary Table 1. The first five genes converged on the primary MAGeCKrs fitting path, while the Legacy MAGeCK mechanism driving their extreme estimates remains unresolved. Conversely, CCDC12 converged via a half-step retry in MAGeCKrs, showing that step-halving retries do not account for the primary-fit outliers or explain all cross-engine differences. Explicit convergence checks ensure numerical auditability but do not establish biological truth.

5.5 Within-engine repeatability

We evaluated run-to-run variation by verifying repeatability within each implementation. Every PC-9 RRA MAGeCKrs run matched its serial reference guide and gene outputs, and every formal PC-9 MLE and HCC827 MLE MAGeCKrs run matched the validated MLE references. MAGeCKrs used zero same-model retries in PC-9 MLE and 15 retries in HCC827 MLE (including CCDC12). Legacy MAGeCK PC-9 MLE runs matched serial qualification at printed beta precision, and Legacy MAGeCK HCC827 MLE outputs were byte-identical across qualification, warmup, and formal runs. Both tools were repeatable under the tested configurations, indicating that the observed cross-engine differences were reproducible rather than caused by run-to-run variation.

6 Discussion

MAGeCKrs integrates the core CRISPR screen analysis stages with explicit numerical verification and inspectable outputs. More broadly, our evaluation shows why benchmarking scientific software requires separating four distinct questions: input identity, workflow throughput, numerical convergence, and inferential agreement.

First, byte-identical guide counts and shared adapter filtering establish a common input foundation for downstream comparison. Even from identical count matrices, however, pipeline conventions can alter downstream tables. In PC-9 RRA, MAGeCKrs retained one Control row and 355 all-zero genes that Legacy MAGeCK omitted, while substantial top-rank overlap coexisted with notable differences in continuous p-values and FDRs. Cross-tool comparisons must therefore distinguish output-retention rules from biological discoveries and recognize that stable top rankings do not imply identical significance thresholds.

Second, timing complete counting-to-analysis workflows captures practical bottlenecks omitted by analysis-only benchmarks. In PC-9 RRA, serial counting across 20 FASTQ files dominated total runtime, whereas the smaller eight-file MLE workflows shifted a larger share of elapsed time onto

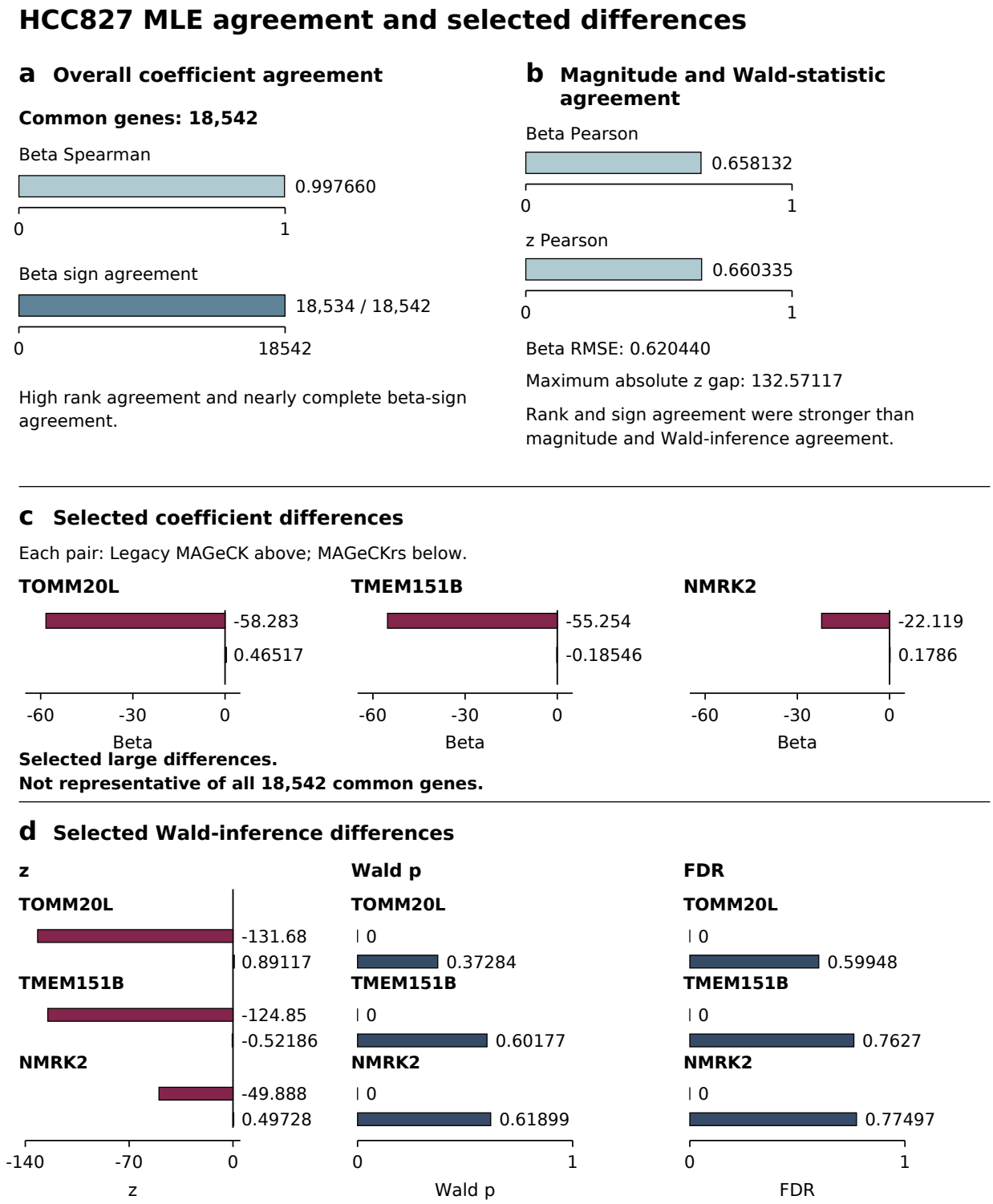


Figure 4. HCC827 MLE agreement and selected differences. (a) Across all 18,542 common genes, Legacy MAGeCK and MAGeCKrs showed high beta rank correlation (Spearman 0.997660) and matching beta signs for 18,534 genes. (b) Linear magnitude and Wald agreement were lower (beta Pearson 0.658132, beta RMSE 0.620440, z Pearson 0.660335, maximum absolute z gap 132.57117). (c, d) Treatment beta estimates (c) and Wald z, p, and FDR values (d) for TOMM20L, TMEM151B, and NMRK2, which are selected examples with large cross-engine differences and are not representative of all 18,542 common genes. Printed zero p-values and FDRs are shown on a linear scale without a logarithmic floor and should not be interpreted as zero biological uncertainty. The complete seven-gene panel appears in Supplementary Table 1; these unresolved differences do not establish which implementation is biologically correct.

model fitting. The 3.65- to 3.77-fold speedups observed here describe end-to-end execution on fixed datasets and hardware; they cannot be attributed to a single statistical kernel or extrapolated to other environments.

Third, cross-implementation agreement depends on the dataset and must be interpreted across both the full gene universe and selected outliers. Whereas PC-9 MLE showed close beta agreement across all 18,542 genes, HCC827 MLE combined high genome-wide rank and sign agreement (Spearman 0.997660; 18,534/18,542 matching signs) with large coefficient and Wald-inference differences in selected genes such as TOMM20L, TMEM151B, and NMRK2. Reporting only aggregate correlations would obscure those outliers, whereas highlighting only the outliers would misrepresent the strong agreement across the rest of the library. Finally, while MAGeCKrs enforces strict convergence checks and logs retry diagnostics, numerical auditability does not establish biological truth. Resolving the Legacy MAGeCK outlier mechanism and establishing biological accuracy will require independent experimental validation and broader calibration studies.

7 Methods

PC-9 RRA, PC-9 MLE, and HCC827 MLE evaluated MAGeCKrs using the binaries compiled for their respective benchmark campaigns. The core scientific implementation governing counting, RRA, and scalar MLE did not change between those campaign builds and the current release source. Campaign-specific executable identities are provided in the supplement, and all reported measurements apply to the evaluated executables and fixed workflows.

7.1 Counting and rank aggregation

Physical FASTQ splits were summed within sample groups. Both engines used serial fixed-offset counting without COUNT normalization, followed by a shared eight-worker canonical adapter that enforced fixed guide eligibility and manifest-order serialization. Because the adapter is part of the benchmarked workflow, its runtime is included in all complete-workflow timings (Supplementary Methods).

RRA converted directional negative-binomial guide statistics into ranked evidence and aggregated them to genes [1, 3]. Multi-control modeling—used in PC-9 RRA—estimated dispersion from unbiased control variance, an ordinary-least-squares trend, and the maximum of raw and fitted dispersion. (A separate single-control procedure uses pooled moments, centered weighted least squares, and fitted-only dispersion, and was not used in PC-9 RRA.) Empirical gene probabilities received separate directional Benjamini-Hochberg adjustment [8], while genes without valid guides received sentinel values of 1.0 outside the correction family. These probabilities are nominal rank-model diagnostics rather than universal guarantees of biological false-discovery calibration.

7.2 MLE and inference

Per-gene negative-binomial regression used the log-link mean

$$\mu_i = s_i \exp(X_i \beta),$$

where s_i is the sample normalization factor and X contains guide-baseline and condition-effect columns. Validated raw counts were incremented by one before fitting, while normalization factors were computed separately. Under the default scalar model (`genes-varmodeling 0`), parameters were estimated via iteratively reweighted least squares. Convergence checks covered all guide and condition coefficients alongside the scaled fixed-point residual at the returned vector. Reduced-relaxation retries re-estimated the same model from the original initialization.

Ordinary Wald statistics were rebuilt at the returned coefficient vector, and unrounded p-values were adjusted per condition across eligible targeting genes via Benjamini-Hochberg [8]. Unresolved planned hypotheses exited with diagnostics rather than writing partial primary summaries, and placeholder permutation columns were excluded from comparisons. Tolerances, retry rules, and exclusions appear in Supplementary Methods.

7.3 Complete-workflow timing and comparisons

Complete-workflow wall time began immediately before COUNT process creation and ended after final output closure. It included process startup and imports, gzip decompression, counting, the canonical adapter, intermediate I/O, stage transitions, statistical analysis and child processes, serialization, and closure. Authentication, staging, directory setup, post-run validation, hashing, reporting, and inter-slot settling were excluded.

For paired Legacy MAGeCK (L_i) and MAGeCKrs (R_i) durations, PC-9 RRA used $\text{median}_i(L_i/R_i)$, whereas PC-9 MLE and HCC827 MLE used $\text{median}_i(L_i)/\,\text{median}_i(R_i)$. All summaries and agreement statistics were transcribed from benchmark records without recomputation. Reported ranges are observed minima and maxima rather than confidence intervals, and PC-9 RRA stage medians are not additive. Output comparisons distinguish native tables from shared common-gene universes.

8 Limitations

Our findings are bounded by the evaluated datasets, hardware, and software configurations. All three workflows derive from one screen collection (E-MTAB-13979) on one Apple Silicon host; small sample sizes ($n = 3$ pairs for PC-9 RRA; $n = 6$ runs per engine for MLE), cache state, and measurement order limit generalization. PC-9 RRA and PC-9 MLE ingest different complete FASTQ inputs, and HCC827 MLE combined high aggregate rank and sign agreement with selected large coefficient and Wald-inference differences. These observations do not establish universal speedup, biological calibration, or identical gene outputs. Partial CPU telemetry and non-uniform memory sampling do not support energy or comparative memory claims, and build qualifications do not transfer to untested binaries. The evaluated implementation supports Apple Silicon macOS; cross-platform behavior and precompiled binary distribution were outside this study.

9 Code availability

MAGeCKrs source code and documentation are available from the MAGeCKrs GitHub repository, `https://github.com/Yoonecorn/MAGeCKrs`, under the repository's MIT license. The release version is 0.1.0 and the release tag is `v0.1.0`. This is a source-only release: no compiled executable is distributed. The supported release target is Apple Silicon macOS. Cross-platform behavior and precompiled binary distribution were outside the evaluated release scope. Building this source-only release does not require an Apple Developer ID, signing, or notarization. Repository access is currently private.

10 Data and evidence availability

Public datasets are available from their original repositories under the accessions reported in this manuscript, including ArrayExpress accession E-MTAB-13979 [5]. The complete sample-to-run mapping is reproduced in the supplement from the benchmark manifests; direct run-level verification against public repository metadata was unavailable when those mappings were recorded, and no new assignment is inferred here.

The MAGeCKrs source code and documentation intended for public use are available through the GitHub repository identified above, subject to its current private access. Detailed internal benchmark execution records, system logs, validation receipts, and local operational evidence are retained privately and are not publicly deposited. Numerical values used in the manuscript, figures, and tables are reproduced in the manuscript or Supplementary Information where applicable. Local filesystem paths are omitted from reader-facing text. No evidence archive or archival DOI is asserted.

11 Author contributions

Yoon Sing Yap conceived the project, designed and implemented the software, developed the methods, performed the validation and formal analyses, curated the data and evidence, prepared the figures, and wrote and revised the manuscript.

12 Competing interests

The author declares no competing interests.

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13 Table legends

Table 1: Workflow and dataset definitions.

Workflow	Campaign and analysis	Compressed FASTQs / sample groups	Compressed bytes	Contrast
PC-9 RRA	Supplemental campaign; COUNT → canonical adapter → RRA/test	20 / 10	10,941,806,594	PC-9 Day 10; osimertinib 7.5 nM R1/R2 versus DMSO R1/R2
PC-9 MLE	V2 campaign; COUNT → canonical adapter → MLE	8 / 4	4,428,475,706	PC-9 Day 10; osimertinib 7.5 nM R1/R2 versus DMSO R1/R2
HCC827 MLE	V2 campaign; COUNT → canonical adapter → MLE	8 / 4	4,351,811,768	HCC827 Day 23; osimertinib 10 nM R1/R2 versus DMSO R1/R2

Internal benchmark identifiers W1, W2, and W3 are retained in the supplement for provenance.

Physical FASTQ splits are summed into sample columns and do not represent independent biological replicates. All workflows derive from E-MTAB-13979 [5]. PC-9 RRA and PC-9 MLE evaluate the same biological contrast from different complete counted inputs, with R1 and R2 denoting the two biological replicates per condition. Source: evidence map C05.

Table 2: Complete-workflow runtime results (seconds).

Workflow / engine	Valid n	Median seconds	Minimum–maximum seconds	Primary runtime ratio
PC-9 RRA / Legacy MAGeCK	3	623.096344	622.315186–628.871025	3.769137010; median paired
PC-9 RRA / MAGeCKrs	3	165.108136	164.099096–168.485985	Same three pairs
PC-9 MLE / Legacy MAGeCK	6	241.969656	238.120982–277.684403	3.670967319; ratio of medians
PC-9 MLE / MAGeCKrs	6	65.914413	65.237344–72.727992	Same six pairs
HCC827 MLE / Legacy MAGeCK	6	237.9524685	234.497993–275.577570	3.654647721; ratio of medians, different scientific outputs
HCC827 MLE / MAGeCKrs	6	65.109550	63.792451–70.263415	Same six pairs; nonidentical HCC827 MLE outputs

All values retain full benchmark precision. PC-9 RRA reports the median within-pair runtime ratio between Legacy MAGeCK and MAGeCKrs, whereas PC-9 MLE and HCC827 MLE report the ratio of engine medians. For HCC827 MLE, the timing ratio reflects completion of a fixed workflow that produced nonidentical gene-level estimates (detailed in Results, Figure 4, and Supplementary Table 1). Minima and maxima represent descriptive observed ranges rather than confidence intervals, and observations are not pooled across campaigns. Sources: C01–C03 and C24.

Table 3: Scientific agreement summary.

Comparison	Accepted result	Interpretation
COUNT, all workflows	0 discrepant integer cells; 114,653 guide IDs	Exact count comparison under fixed matching and grouping
Adapter, all workflows	113,806 retained guides; byte identity to accepted tables	Fixed eligibility and manifest order
PC-9 RRA native gene rows	Legacy MAGeCK 18,187; MAGeCKrs 18,543; common 18,187	One Control plus 355 all-zero retained rows are not discoveries
PC-9 RRA negative rank overlap	82/100; 410/500	Descriptive ranking agreement
PC-9 RRA positive rank overlap	80/100; 397/500	Descriptive ranking agreement
PC-9 MLE beta	Spearman 0.999996; signs 18,542/18,542; maximum absolute gap 0.04862, U2AF2	Close but nonidentical estimates
HCC827 MLE beta	Spearman 0.997660; Pearson 0.658132; RMSE 0.620440; signs 18,534/18,542	High beta-rank and sign agreement with selected material coefficient differences; complete selected panel in Supplementary Table 1
HCC827 MLE z	Pearson 0.660335; maximum absolute gap 132.57117	Selected material Wald-inference differences; unresolved mechanism
MLE permutation-style columns	Legacy MAGeCK nan; MAGeCKrs 1.0	Excluded from inferential equivalence

Exact count and adapter validation precedes all gene-level comparisons, which distinguish native table outputs from shared common-gene universes. Retained control or all-zero rows in RRA and placeholder permutation columns in MLE do not represent biological discoveries or empirical permutation inference. The complete seven-gene HCC827 MLE difference panel and first-three-gene FDR values are provided in Supplementary Table 1; reported coefficient gaps are transcribed directly from the benchmark records rather than recomputed from rounded display values. Sources: C06–C14 and C22.

Supplementary information

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S1 Supplementary Methods

The internal benchmark records identify the PC-9 RRA, PC-9 MLE, and HCC827 MLE workflows as W1, W2, and W3, respectively.

S1.1 COUNT and canonical adapter

COUNT accepts space-separated sample groups and comma-separated physical FASTQ files within each group, summing counts across physical splits into the group's output column. Both engines used serial fixed-offset counting with `trim-5 23`, guide length 19, and no COUNT normalization. MAGeCKrs used exact matching (mismatch setting 0) and one COUNT worker. Its general matching interface also permits unique single-mismatch assignment at fixed offsets while skipping ambiguous matches; automatic offset scanning differs from Legacy MAGeCK and was not used here.

A shared eight-worker canonical adapter processed both engines' freshly counted tables and serialized the output in manifest order. Guide eligibility and negative-control identifiers were fixed before execution. Ineligible guides and fully lost targets were excluded without inserting zero-filled rows. The adapter is benchmark infrastructure included in the timing boundary rather than a standalone `mageckrs` command.

S1.2 RRA and negative-binomial statistics

Multi-control RRA—used for PC-9 RRA—estimates guide dispersion using unbiased control variance, an ordinary-least-squares trend, and the maximum of raw and fitted dispersion. A separate single-control adapter path uses pooled control/treatment moments, centered weighted least squares, and fitted-only dispersion at the control mean; that approximation was not used in PC-9 RRA.

Directional valid-guide statistics are converted to global mid-rank fractions in $(0, 1]$. For a gene with k valid guides before alpha truncation, the RRA lo-value is the minimum of $F_{\mathrm{Beta}(j,k-j+1)}(r_j)$ across retained sorted ranks [1, 3]. If alpha retains no rank, the smallest rank is scored. A gene with no valid guides receives score, p-value, and FDR sentinels of 1.0 and is excluded from that directional Benjamini–Hochberg family [8].

The default guide threshold is 0.25, which sets directional alpha rather than filtering output genes by FDR. Passing fractions are clamped to $[0.05, 0.5]$, with a positive-direction fallback of 0.01 when no positive guide passes. Null sampling uses 1,000 draws per required valid-guide-count stratum per direction by default. Standard nulls draw open-uniform rank fractions, whereas control-guided nulls sample with replacement from directional control-percentile pools (failing if the valid pool is empty). Empirical p-values use nearest-index lookups within $\pm 10^{-9}$ of the observed score as $p_{\mathrm{emp}} = (\mathrm{left} + \mathrm{right} + 1)/(2P)$, where P is the per-stratum draw count, followed by separate directional Benjamini–Hochberg adjustment [8].

PC-9 RRA used control normalization across 1,004 canonical controls, `remove-zero both` at threshold 0, guide threshold 0.25, and FDR adjustment. MAGeCKrs used eight CPU workers, 1,000 permutations, and seed 0 (distinct from the general default seed 81985529216486895), whereas Legacy MAGeCK used its native single-process RRA binary. RRA p-values and FDRs are nominal

rank-model diagnostics that require interpretation alongside effect direction, guide consistency, and experimental replication.

S1.3 MLE model, convergence and inference

MLE aligns sample columns with the design matrix and verifies full column rank before fitting. Per-gene negative-binomial regression uses the log-link mean $\mu_i = s_i \exp(X_i\beta)$, where s_i is the sample normalization factor expanded to observation i and X contains guide-baseline and condition-effect columns. Validated raw counts are incremented by 1 before fitting, while normalization factors are computed separately. By default, guide efficiencies are initialized to 1 without iterative updates, and `genes-varmodeling 0` uses scalar dispersion $\alpha = 0.01$. Genes exceeding the default limit of 500 guides are skipped in final fitting but retained in normalization with unit efficiency. Fitting uses iteratively reweighted least squares (IRLS).

Under the MAGeCKrs numerical convergence contract, every guide and condition coefficient must satisfy absolute tolerance 10^{-8} and relative tolerance 10^{-6} across three consecutive finite updates, with an accepted-point scaled fixed-point residual of at most 10^{-9}, coefficient magnitude at most 100, and at most 1,000 candidate requests per attempt. The residual checks IRLS fixed-point stability rather than proving a global optimum. Eligible non-converged fits retry the same model with relaxation 1, 0.5, and 0.25 from the original validated initialization, keeping dispersion and ridge fixed without clipping, warm starts, or model inflation. Retry attempts and exclusions are logged in diagnostics.

Final means, weights, normal-equation inverses, and scaled sandwich covariance matrices are rebuilt at the returned coefficient vector. Valid Wald inference requires strictly positive finite standard errors and valid z statistics and p-values; degenerate standard errors are never replaced with arbitrary large constants. Per-condition Benjamini–Hochberg adjustment [8] uses unrounded ordinary Wald p-values across the shared eligible targeting-gene universe, excluding control, declared all-control, pre-skipped, and failed genes. An unresolved planned fit or Wald hypothesis—or an empty eligible universe—exits with nonzero status and diagnostic logs rather than writing incomplete primary summaries.

PC-9 MLE and HCC827 MLE used control normalization, `genes-varmodeling 0`, and eight MLE workers (threads in MAGeCKrs versus compute processes with manager/controller queues in Legacy MAGeCK). Legacy MAGeCK permutation-round was 0 and max-sgrnapergene-permutation was 501; MAGeCKrs maximum guides per gene was 500. Comparisons cover treatment betas and ordinary Wald statistics only; permutation p/FDR columns were `nan` in Legacy MAGeCK and `1.0` placeholders in MAGeCKrs.

S1.4 Output auditability and publication behavior

Tab-separated summaries preserve input sample order and pipe-delimited condition columns, formatting normalized counts to four decimal places. RRA guide-summary significance fields are omitted, and MLE permutation-style fields are placeholders. RRA metadata record the effective seed, while MLE diagnostics log retry attempts, exclusions, and inference populations.

Staged workflows publish output sets via sequential atomic per-file renames with best-effort rollback, rejecting pre-existing targets (standalone COUNT retains overwrite behavior). Any unresolved planned fit or inference failure aborts primary summary output and writes diagnostics.

S2 Exact benchmark configurations

All workflows used serial fixed-offset COUNT with `trim-5 23`, guide length 19, and no COUNT normalization. MAGeCKrs used exact matching (mismatch 0) and one COUNT worker, followed by a shared eight-worker canonical adapter that serialized results in manifest order. These fixed worker counts define the benchmark protocol rather than a scaling study.

PC-9 RRA used control normalization across 1,004 canonical controls, `remove-zero both` at threshold 0, guide threshold 0.25, and FDR adjustment. MAGeCKrs used eight CPU workers, 1,000 permutations, and seed 0 (distinct from the general default seed 81985529216486895), whereas Legacy MAGeCK used native RRA without an artificial worker flag.

PC-9 MLE and HCC827 MLE used control normalization, `genes-varmodeling 0`, and eight MLE workers per engine (threads in MAGeCKrs versus compute processes in Legacy MAGeCK). Legacy MAGeCK permutation-round was 0 and max-sgrnapergene-permutation was 501; MAGeCKrs maximum guides per gene was 500. Comparisons evaluate beta and ordinary Wald outputs only, excluding permutation p/FDR placeholders (`nan` in Legacy MAGeCK and `1.0` in MAGeCKrs).

OMP, OpenBLAS, MKL, VECLIB, and NUMEXPR thread counts were pinned to one, keeping nested linear-algebra threads separate from analysis worker counts. No automatic worker tuning, CPU affinity/QoS overrides, or Metal execution entered formal measurements. Locale and environment bindings were fixed within each campaign, and exact binary identities appear in Supplementary Table 4.

S3 Exact timing definitions

Complete-workflow wall time began immediately before COUNT process creation and ended after final output closure. It included process startup and imports, gzip decompression, COUNT, the canonical adapter, intermediate I/O, stage transitions, statistical analysis (including native child processes and wrapper overhead in Legacy MAGeCK RRA), serialization, and closure.

Authentication, staging, output-directory setup, post-run validation, hashing, reporting, and inter-slot settling (at least 30 seconds) were excluded. Cache state and shared-host activity were not experimentally isolated, and post-run cleanup in PC-9 RRA occurred outside primary time. Stage medians in PC-9 RRA are summarized independently and do not sum to complete-workflow time.

PC-9 RRA comprised three fresh valid pairs in Legacy MAGeCK/MAGeCKrs, MAGeCKrs/Legacy MAGeCK, and Legacy MAGeCK/MAGeCKrs order, without dataset warmup, scouting, tuning, or replacement. Its estimator was $\text{median}_i(L_i/R_i)$. The V2 campaign supplied six valid observations per engine for each of PC-9 MLE and HCC827 MLE, with estimator $\text{median}_i(L_i)/\,\text{median}_i(R_i)$, excluding qualification and warmup runs. Campaigns are never pooled, and reported minima and maxima are observed ranges rather than confidence intervals or interquartile ranges.

The PC-9 RRA 21,600-second limit was a whole-slot safety ceiling never reached by valid runs. An earlier historical 7,200-second failure resulted from a space-containing control-file path defect rather than working RRA runtime, and failed historical runs were excluded without substitution. V2 updated its supervisor script after the second formal round; the timing boundary remained equivalent apart from small symmetric wrapper overhead.

Exact accepted complete-workflow summaries.

Workflow / campaign	Engine	Valid n / planned	Median s	Minimum s	Maximum s
PC-9 RRA / supplement	Legacy MAGeCK	3 / 3	623.096344	622.315186	628.871025

Workflow / campaign	Engine	Valid n / planned	Median s	Minimum s	Maximum s
PC-9 RRA / supplement	MAGeCKrs	3 / 3	165.108136	164.099096	168.485985
PC-9 MLE / V2	Legacy MAGeCK	6 / 6	241.969656	238.120982	277.684403
PC-9 MLE / V2	MAGeCKrs	6 / 6	65.914413	65.237344	72.727992
HCC827 MLE / V2; selected-difference panel below	Legacy MAGeCK	6 / 6	237.9524685	234.497993	275.577570
HCC827 MLE / V2; selected-difference panel below	MAGeCKrs	6 / 6	65.109550	63.792451	70.263415

HCC827 MLE timings reflect completion of a fixed workflow across engines that produced nonidentical estimates on selected genes (Supplementary Table 1). All values above are transcribed without recomputation.

S4 Individual timing observations

Table S2: Individual complete-workflow timing observations.

Campaign / workflow	Pair	Legacy MAGeCK s	MAGeCKrs s	Paired runtime ratio
Supplement / PC-9 RRA	P01	628.871025	168.485985	3.732482705
Supplement / PC-9 RRA	P02	623.096344	164.099096	3.797073593
Supplement / PC-9 RRA	P03	622.315186	165.108136	3.769137010
V2 / PC-9 MLE	M01	266.594784	69.320332	3.845838246
V2 / PC-9 MLE	M02	277.684403	72.727992	3.818122780
V2 / PC-9 MLE	M03	238.120982	65.597290	3.630042979
V2 / PC-9 MLE	M04	242.042973	66.214367	3.655444943
V2 / PC-9 MLE	M05	241.896339	65.237344	3.707942785
V2 / PC-9 MLE	M06	238.267920	65.614459	3.631332539
V2 / HCC827 MLE	M01	265.202528	69.323340	3.825587861
V2 / HCC827 MLE	M02	275.577570	70.263415	3.922063424
V2 / HCC827 MLE	M03	234.528139	64.405720	3.641417859
V2 / HCC827 MLE	M04	237.006502	64.377208	3.681528127
V2 / HCC827 MLE	M05	234.497993	65.813380	3.563074758
V2 / HCC827 MLE	M06	238.898435	63.792451	3.744932688

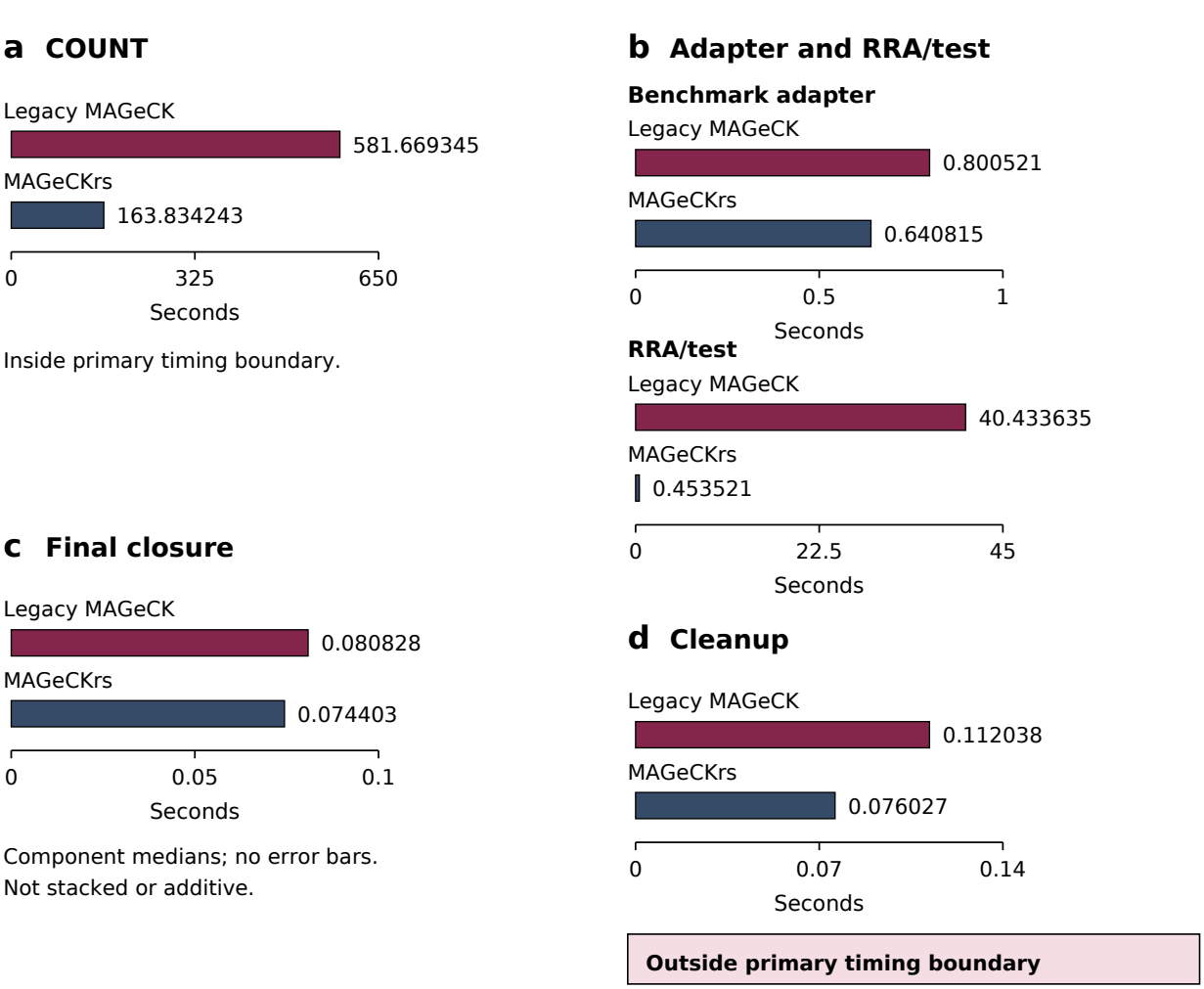
All 15 paired rounds (30 measured durations) and within-pair ratios are transcribed from the retained benchmark records. Primary estimators remain the median paired ratio for PC-9 RRA and the ratio of engine medians for PC-9 MLE and HCC827 MLE.

S5 Complete PC-9 RRA stage summary

Table S3: Complete PC-9 RRA stage summary (median seconds).

PC-9 RRA component	Legacy MAGeCK seconds	MAGeCKrs seconds	Timing boundary
COUNT	581.669345	163.834243	Inside primary
Canonical adapter	0.800521	0.640815	Inside primary
RRA/test	40.433635	0.453521	Inside primary
Final closure	0.080828	0.074403	Inside primary
Cleanup	0.112038	0.076027	Outside primary

Each stage duration is summarized by its own median across the three PC-9 RRA rounds. The RRA/`test` stage encompasses Python guide-level modeling and both directional calls to the native RRA binary in Legacy MAGeCK. Individual stage ratios do not represent complete-workflow speedups, and post-run cleanup is excluded from the primary timing boundary.



Supplementary Figure 1. PC-9 RRA stage timings. Median elapsed times for Legacy MAGeCK and MAGeCKrs across (a) serial COUNT, (b) the canonical benchmark adapter and RRA/`test`, (c) final output closure, and (d) post-run cleanup (outside the primary timing boundary). Horizontal bars display independently summarized stage medians on stage-specific scales rather than stacked additive totals; no error bars are shown because stage-level ranges were not separately tabulated. Source values are listed in Supplementary Table 3.

S6 HCC827 MLE seven-gene selected-difference table

Across 18,542 common genes, beta Spearman correlation was 0.997660 and beta signs agreed for 18,534/18,542. Beta Pearson correlation was 0.658132 and RMSE was 0.620440. For z statistics, Pearson correlation was 0.660335 and the maximum absolute difference was 132.57117. These aggregate metrics should be interpreted with the selected large coefficient and Wald-inference differences below; the selected genes are not representative of all common genes.

Table S1: HCC827 MLE seven-gene selected-difference table.

Gene	Legacy MAGeCK beta	MAGeCKrs beta	Absolute beta gap	Legacy MAGeCK z / Wald p	MAGeCKrs z / Wald p	Classification
TOMM20L	-58.283	0.46517	58.74817	-131.68 / 0	0.89117 / 0.37284	Primary-fit discrepancy; Legacy MAGeCK mechanism unresolved
TMEM151B	-55.254	-0.18546	55.06854	-124.85 / 0	-0.52186 / 0.60177	Primary-fit discrepancy; Legacy MAGeCK mechanism unresolved
NMRK2	-22.119	0.1786	22.29760	-49.888 / 0	0.49728 / 0.61899	Primary-fit discrepancy; Legacy MAGeCK mechanism unresolved
MAP2K1	-9.4826	-0.41615	9.06645	-21.438 / 5.9296e-102	-1.5073 / 0.13173	Primary-fit discrepancy; Legacy MAGeCK mechanism unresolved
LMO4	-5.1646	-0.015783	5.14882	-11.681 / 1.5986e-31	-0.060924 / 0.95142	Primary-fit discrepancy; Legacy MAGeCK mechanism unresolved
CCDC12	3.7028	1.1286	2.57420	0.06905 / 0.94495	1.0199 / 0.30778	Accepted half-step retry example; does not explain the primary-fit outliers or all gaps
RPL36	0.28047	2.9897	2.70923	1.2255 / 0.22038	0.40651 / 0.68437	Largest retried-gene beta gap

See numerical evidence map, C10–C12. Accepted gaps are preserved, not recalculated. HCC827 MLE is not a same-answer speedup.
First-three-gene Legacy MAGeCK Wald p/FDR: 0/0. MAGeCKrs p/FDR: TOMM20L 0.37284/0.59948; TMEM151B 0.60177/0.7627; NMRK2 0.61899/0.77497.
Primary-fit Legacy MAGeCK mechanism remains unresolved. Corrected convergence does not establish biological truth.

In Supplementary Table 1, the first five genes (TOMM20L, TMEM151B, NMRK2, MAP2K1, and LMO4) represent primary-fit divergences: each converged directly on the primary IRLS path in MAGeCKrs, whereas the numerical mechanism producing extreme negative betas in Legacy MAGeCK remains unresolved. CCDC12 illustrates a gene that converged via a half-step relaxation retry in MAGeCKrs; however, step-halving retries do not account for the five primary-fit outliers or explain all cross-engine differences. RPL36 exhibited the largest absolute beta difference among retried genes. All reported beta gaps are transcribed directly from the benchmark comparison records rather than recomputed from rounded display values.

For TOMM20L, TMEM151B, and NMRK2, Legacy MAGeCK reported printed Wald p-values and FDRs of 0/0, whereas MAGeCKrs reported p/FDR values of 0.37284/0.59948, 0.60177/0.7627, and 0.61899/0.77497, respectively. Legacy MAGeCK printed these p-values and FDRs as zero; the printed zeros do not imply zero biological uncertainty. FDR values for the remaining four supplementary genes were not separately tabulated in the primary record.

Across all 18,542 common genes in HCC827 MLE, treatment betas had a Spearman correlation of 0.997660, a Pearson correlation of 0.658132, an RMSE of 0.620440, and matching signs for 18,534/18,542 genes, while Wald z statistics had a Pearson correlation of 0.660335 and a maximum absolute difference of 132.57117. As noted in the main text, numerical convergence in MAGeCKrs ensures auditability but does not establish biological truth.

S7 PC-9 RRA native and common output-universe details

Raw COUNT validation covered all 114,653 canonical guide IDs with zero discrepant integer cells across 1,146,530 cells in PC-9 RRA and 458,612 cells in each of PC-9 MLE and HCC827 MLE. The shared canonical adapter retained 113,806 guides (112,802 targeting guides and 1,004 negative controls), excluded 847 ineligible guides, and omitted 200 targets with no remaining guides rather than zero-filling them. Every formal benchmark run produced a filtered count table that matched its reference downstream input byte for byte.

In PC-9 RRA, Legacy MAGeCK emitted 18,187 native gene rows and MAGeCKrs emitted 18,543 rows, sharing a common universe of 18,187 targeting genes. The 356 additional rows in MAGeCKrs consist of one aggregated Control row and 355 targeting genes with zero counts across the four Day 10 contrast samples; these retained rows are distinct from the 200 targets removed by the adapter and do not represent biological discoveries. Top-ranked gene overlaps were 82/100 and 410/500 in the negative direction and 80/100 and 397/500 in the positive direction under both native and common-universe ranking, while continuous RRA scores, p-values, FDRs, ranks, and log fold changes differed across engines.

PC-9 RRA common-universe maximum absolute printed-value differences.

Field	Negative direction	Positive direction
Score	0.73274426	0.76114396
p-value	0.72155	0.811954
FDR	0.945401	0.789528212
Rank	9,517	9,474
LFC	0.84928	0.84928

The maximum absolute differences above were evaluated across the 18,187 common targeting genes using printed output values (see numerical evidence map, C13). Because each summary field operates on a different numerical scale, high overlap among top-ranked genes does not imply identical FDR calibration or exact numerical equivalence.

S8 Complete accession and sample mapping

Sample-to-run mapping from the accepted manifests.

Sample label	FASTQ accessions	Workflow
PC-9-Yusa-DMSO-d3-R1	ERR12815343, ERR12815344	PC-9 RRA
PC-9-Yusa-DMSO-d3-R2	ERR12815345, ERR12815346	PC-9 RRA
PC-9-Yusa-DMSO-d10-R1	ERR12815351, ERR12815352	PC-9 RRA, PC-9 MLE
PC-9-Yusa-DMSO-d10-R2	ERR12815353, ERR12815354	PC-9 RRA, PC-9 MLE
PC-9-Yusa-Osi-7p5nM-d10-R1	ERR12815355, ERR12815356	PC-9 RRA, PC-9 MLE
PC-9-Yusa-Osi-7p5nM-d10-R2	ERR12815359, ERR12815360	PC-9 RRA, PC-9 MLE
PC-9-Yusa-DMSO-d17-R1	ERR12815365, ERR12815366	PC-9 RRA
PC-9-Yusa-DMSO-d17-R2	ERR12815367, ERR12815368	PC-9 RRA
PC-9-Yusa-Osi-7p5nM-d17-R1	ERR12815369, ERR12815370	PC-9 RRA
PC-9-Yusa-Osi-7p5nM-d17-R2	ERR12815371, ERR12815372	PC-9 RRA
HCC827-Yusa-DMSO-d23-R1	ERR12815325, ERR12815326	HCC827 MLE
HCC827-Yusa-DMSO-d23-R2	ERR12815327, ERR12815328	HCC827 MLE
HCC827-Yusa-Osi-10nM-d23-R1	ERR12815329, ERR12815330	HCC827 MLE
HCC827-Yusa-Osi-10nM-d23-R2	ERR12815331, ERR12815332	HCC827 MLE

Sample labels and paired FASTQ run accessions are transcribed from the benchmark manifests (see numerical evidence map, C05 and C25) for ArrayExpress study E-MTAB-13979 [5]. Because direct run-level cross-checking against external archive metadata was unavailable when these mappings were recorded, these mappings reflect the benchmark manifests without inferring new run assignments. Paired ERR files represent technical sequencing splits that are summed within each sample column rather than independent biological replicates.

S9 Hardware and software identities

Table S4: Hardware and software identities.

Item	Accepted recorded identity	Qualification boundary
Benchmark hardware	Mac mini Mac16,11; Apple M4 Pro; 12 CPU cores; 24 GB RAM	Single recorded host; PC-9 RRA did not freshly attest chip/RAM
Operating system	macOS 27.0.1, build 26A434; arm64; APFS	No older-macOS or other-chip qualification
PC-9 RRA MAGeCKrs executable	mageckrs 0.1.0; executable identity below	Accepted native-CPU benchmark binary
PC-9 MLE/HCC827 MLE MAGeCKrs executable	mageck2-rs 0.1.0; executable identity below	Campaign-specific executable
Legacy MAGeCK environment	MAGeCK 0.5.9.5; Python 3.12.15; NumPy 2.5.3; SciPy 1.18.1; native RRA 0.5.9	Recorded comparator environment
Legacy MAGeCK linear algebra	V2 conda libblas/liblapack 3.9.0	No assumed Accelerate runtime
MAGeCKrs build	rustc 1.98.0; LLVM 22.1.8; PC-9 RRA native CPU, fat LTO, one codegen unit, panic abort, stripped symbols	Default Accelerate; no Metal; dynamic system libraries
Separate technical qualification	Same-host generic ARM64 builds from the accepted source	Different binary; no transferred benchmark timing

Campaign-specific executable identities.

Item	PC-9 RRA supplement	PC-9 MLE/HCC827 MLE V2 campaign
MAGeCKrs executable name/version	mageckrs 0.1.0	mageck2-rs 0.1.0
Binary bytes	2,588,624	2,588,640
Binary SHA-256	`7bd60fc40d46621a9a6c5eda903eaf7135fbe0c07a791378db291cb5c04cd1bb`	`a788853328069ae101e73dc2401c9a5e7af5fb1d2f702a288d501198022d74cb`

The benchmark host had 25,769,803,776 bytes (24 GB) of RAM, and the V2 campaign recorded Darwin kernel version 27.0.0. Hardware chip and memory identifiers were captured during the V2 campaign and were not re-queried during the PC-9 RRA supplemental campaign. The PC-9 RRA campaign executed the pre-built native-CPU `mageckrs 0.1.0` binary directly, whereas the V2 MLE campaign executed the campaign-specific `mageck2-rs 0.1.0` binary; runtime measurements apply strictly to those exact binaries.

In a separate build-qualification check on the same host, the release test suite passed all 1,094 all-target tests with zero failures or skipped tests, alongside two separately executed end-to-end COUNT fixture checks and one documentation test (which do not add new biological benchmark datasets). Two clean generic ARM64 builds compiled from the accepted source tree were byte-identical (2,588,640 bytes; SHA-256 `77b5b794a02faedd5fada61388b084de06d542f0e29586bc30f121edbb802680`). Because this generic qualification binary differs from the campaign-specific benchmark executables, no runtime performance claims are transferred to it.

S10 Supported, unsupported, and deferred scope

Table S5: Supported, unsupported and deferred scope.

Area	Current disposition	Claim boundary
COUNT, RRA/test, default scalar MLE	Implemented; bounded technical and accepted workflow evidence	Present manuscript core
Fixed CPU workers	Deterministic tested gene-fit/RRA scheduling; COUNT stays serial	No scaling curve or universal optimal worker count
Exact UMI, COUNT-to-test run, CNV adapter	Documented current core interfaces	Outside the new bounded qualification; no biological UMI/CNV accuracy claim
Automatic worker policy	Available in accepted integration evidence	Not used in formal performance comparisons
MLE permutation inference and MLE seed	Permutation inference deferred; supplied seed rejected	Placeholder fields are not significance estimates
Pathway/GSEA	Reserved pathway command rejects execution	No pathway analysis result
Metal	Compilation evidence only	Operational GPU acceleration unqualified
Platform	macOS Apple Silicon only	No Linux, Windows or Intel Mac support
Distribution	Same-host build/install evidence	Public binary distribution and cross-platform behavior were outside this study
Biological interpretation	Per-engine validation plus disclosed agreement and discrepancies	No universal calibration, identical-output or additional-discovery claim

The evaluated MAGeCKrs implementation targets Apple Silicon macOS; cross-platform support and precompiled binary distribution were outside the scope of this work. While the codebase includes interfaces for exact UMI counting, combined COUNT-to-`test` execution, and post-fit CNV beta adjustment (which shifts matched gene betas without recomputing standard errors or p-values), those modes were not benchmarked here and carry no biological accuracy claims. Similarly, nominal RRA probabilities and ordinary Wald MLE statistics do not guarantee universal false-discovery calibration, and system linear-algebra libraries (including Apple Accelerate) are linked dynamically rather than statically.

S11 Numerical evidence map

This reader-facing map identifies the manuscript locations for numerical claims C01–C27 and their interpretive limits. The numerical values are transcribed from retained benchmark and qualification records without new calculations. Those detailed internal records are private and are not publicly deposited; this map provides internal cross-checks within the manuscript rather than access to the underlying execution records.

S11.1 C01

Manuscript location: Table 2 and Supplementary Table 2.
Accepted values: Legacy MAGeCK 623.096344 s; MAGeCKrs 165.108136 s; 3.769137010 median paired ratio; three valid paired rounds.
Evidence: Accepted formal observations → derived medians/ratio; descriptive n. Limit: Complete fixed workflow, n=3; campaigns not pooled; no interquartile-range inference.

S11.2 C02

Manuscript location: Table 2 and Supplementary Table 2.
Accepted values: 241.969656 / 65.914413 s; 3.670967319 ratio of medians; six observations per engine. Accepted rounded display 241.97 / 65.91; 3.671.
Evidence: Derived from formal observations. Limit: Campaign-specific executable, fixed PC-9 input.

S11.3 C03

Manuscript location: Table 2 and Supplementary Table 1.
Accepted values: 237.9524685 / 65.109550 s; 3.654647721 ratio of medians; six observations per engine. Rounded 237.95 / 65.11; 3.655.
Evidence: Derived from formal observations. Limit: Main three-example disclosure and full Supplementary Table 1; not same-answer speedup.

S11.4 C04

Manuscript location: Supplementary Table 3.
Accepted values: COUNT 581.669345 / 163.834243 s; RRA 40.433635 / 0.453521 s; adapter 0.800521 / 0.640815; closure 0.080828 / 0.074403; cleanup 0.112038 / 0.076027. Rounded COUNT 581.67 / 163.83; RRA 40.43 / 0.454.
Evidence: Derived component medians. Limit: Stages not end-to-end, medians not additive; cleanup outside primary.

S11.5 C05

Manuscript location: Table 1 and Section S8.
Accepted values: PC-9 RRA 20 files / 10 groups / 10,941,806,594 bytes; PC-9 MLE 8 / 4 / 4,428,475,706; HCC827 MLE 8 / 4 / 4,351,811,768. PC-9 Day 10 7.5 nM, HCC827 Day 23 10 nM; R1 / R2.
Evidence: Descriptive input identity. Limit: Splits not biological replicates; PC-9 RRA/PC-9 MLE different counted inputs.

S11.6 C06

Manuscript location: Figure 3 and Section S7.
Accepted values: 114,653 guide IDs; PC-9 RRA 1,146,530 cells; PC-9 MLE / HCC827 MLE 458,612; zero mismatches; 113,806 retained =112,802 targeting+1,004 controls; 847 excluded; 200 lost targets.
Evidence: Validated count observations/descriptive populations. Limit: Canonical eligibility only; do not equate 200 adapter losses with 355 all-zero contrast genes.

S11.7 C07

Manuscript location: Table 3 and Section S7.
Accepted values: Legacy MAGeCK 18,187; MAGeCKrs 18,543; common 18,187; difference 356 = one Control +355 all-zero; MLE 18,542.
Evidence: Validated/descriptive universe. Limit: Retention difference, not discoveries.

S11.8 C08

Manuscript location: Table 3.
Accepted values: Negative 82 / 100 and 410 / 500; positive 80 / 100 and 397 / 500.
Evidence: Accepted derived rank-overlap counts. Limit: Same overlaps native/common; not identical significance or biological calibration.

S11.9 C09

Manuscript location: Table 3.
Accepted values: Spearman 0.999996; signs 18,542 / 18,542; max absolute beta gap 0.04862 U2AF2.
Evidence: Accepted derived agreement. Limit: Rounded accepted metric; no exact equivalence.

S11.10 C10

Manuscript location: Supplementary Table 1.
Accepted values: Every cell in Supplementary Table 1, unchanged.
Evidence: Printed beta/z/p measured output; absolute gap derived in accepted record. Limit: Three examples and aggregate metrics accompany main HCC827 MLE narrative; full panel in Supplementary Table 1; unresolved Legacy MAGeCK mechanism.

S11.11 C11

Manuscript location: Figure 4 and Results.
Accepted values: 18,542 genes; beta Spearman correlation 0.997660, Pearson correlation 0.658132, RMSE 0.620440; Wald z Pearson correlation 0.660335, maximum gap 132.57117; matching signs 18,534 / 18,542.
Evidence: Derived accepted comparisons. Limit: High Spearman does not negate large magnitude/significance differences.

S11.12 C12

Manuscript location: Results and Section S6.
Accepted values: First three Legacy MAGeCK p / FDR 0 / 0; MAGeCKrs 0.37284 / 0.59948; 0.60177 / 0.7627; 0.61899 / 0.77497.
Evidence: Printed native outputs, descriptive. Limit: Preserve printed zeros; do not infer biological certainty.

S11.13 C13

Manuscript location: Section S7.
Accepted values: Negative / positive max absolute score gap 0.73274426 / 0.76114396; p 0.72155 / 0.811954; FDR 0.945401 / 0.789528212; rank 9,517 / 9,474; LFC 0.84928 / 0.84928.
Evidence: Derived accepted printed-value comparison. Limit: 18,187 common genes; magnitude scale differs by field.

S11.14 C14

Manuscript location: Results: within-engine repeatability.
Accepted values: PC-9 MLE zero retries; HCC827 MLE 15; Legacy MAGeCK PC-9 MLE printed beta max gap 0.0000.
Evidence: Validated within-engine observations. Limit: Not cross-engine identity; CCDC12 retry not primary-outlier explanation.

S11.15 C15

Manuscript location: Section S9.
Accepted values: 1,094 passed; zero failed / ignored; two separately invoked end-to-end tests; one documentation test; two fresh byte-identical generic builds.
Evidence: Retained test/build observations. Limit: End-to-end checks were repeated separately and are not extra unique tests; same-host only; no benchmark rerun.

S11.16 C16

Manuscript location: Supplementary Table 4.
Accepted values: Host Mac16,11 / M4 Pro / 12 cores / 24 GB; macOS 27.0.1 build 26A434; mageckrs 0.1.0 vs mageck2-rs 0.1.0; MAGeCK 0.5.9.5 / Python 3.12.15 / NumPy 2.5.3 / SciPy 1.18.1 / RRA 0.5.9; libblas / liblapack 3.9.0; rustc 1.98.0 / LLVM 22.1.8; one codegen unit.
Evidence: Descriptive recorded identities. Limit: Recorded host, PC-9 RRA chip/RAM not freshly attested; no binary performance transfer.

S11.17 C17

Manuscript location: Section S2.
Accepted values: trim 23, guide length 19, normalization none, mismatch 0, COUNT worker 1; adapter workers 8.
Evidence: Descriptive fixed configuration. Limit: Fixed matching; no AUTO equivalence.

S11.18 C18

Manuscript location: Section S2.
Accepted values: 1,004 controls; threshold 0.25; remove-zero threshold 0; workers 8; permutations 1000; benchmark seed 0.
Evidence: Descriptive accepted configuration. Limit: Legacy MAGeCK native RRA lacks equivalent worker flag.

S11.19 C19

Manuscript location: Section S1: RRA.
Accepted values: rank fraction (0,1]; Beta(j,k-j+1); sentinels 1.0; alpha [0.05,0.5], positive fallback 0.01; ±1e-9; (left+right+1)/(2P); default seed 81985529216486895.
Evidence: Descriptive algorithm contract. Limit: No new calibration result; default seed distinct from PC-9 RRA override.

S11.20 C20

Manuscript location: Section S1: MLE.
Accepted values: genes-varmodeling 0; scalar alpha 0.01; maximum guides 500; retained efficiency 1; workers 8; Legacy MAGeCK permutation 0 and max-sgrnapergene-permutation 501.
Evidence: Descriptive configuration. Limit: Model and worker scheduling distinct; beta/ordinary Wald subset.

S11.21 C21

Manuscript location: Section S1: MLE.
Accepted values: absolute tolerance 1e-8 / relative tolerance 1e-6; residual ≤1e-9; three consecutive updates; bound 100; ≤1000 candidate requests; relaxation 1,0.5,0.25.
Evidence: Descriptive accepted numerical contract. Limit: Fixed-point residual, no global-optimum or biological truth claim.

S11.22 C22

Manuscript location: Section S1: output auditability.
Accepted values: normalized counts four decimals; sentinel 1.0; permutation-style MAGeCKrs 1.0 vs Legacy MAGeCK nan.
Evidence: Descriptive output/schema. Limit: Placeholders excluded from inferred significance.

S11.23 C23

Manuscript location: Section S3.
Accepted values: $\geq$30 seconds settle; 21,600 safety ceiling; historical 7,200 defect; V2 transition after second formal round.
Evidence: Descriptive protocol/failure history. Limit: Neither deadline is observed successful RRA runtime.

S11.24 C24

Manuscript location: Table 2 and Supplementary Table 2.
Accepted values: Every per-engine minimum / maximum and every individual observation in Supplementary Table 2.
Evidence: Measured durations; derived ranges. Limit: Range not CI; no pooling or new ratio.

S11.25 C25

Manuscript location: References and Section S8.
Accepted values: All reference years / volumes / pages / DOIs and E-MTAB-13979 / ERR identifiers as recorded.
Evidence: Descriptive bibliographic/identifier metadata. Limit: Public study verified; run-level assignments locally accepted only.

S11.26 C26

Manuscript location: Code availability and Supplementary Table 5.
Accepted values: Native macOS ARM64; MAGeCKrs / mageckrs; version 0.1.0; MIT selected.
Evidence: Release scope. Limit: Source-only release; repository access is currently private; no precompiled executable is distributed.

S11.27 C27

Manuscript location: Methods: MLE and Section S1: MLE. Mean $\mu_i = s_i \exp(X_i\beta)$; guide-baseline and condition columns; raw count plus one; efficiencies initialize to one without iterative updates. These are model definitions, not a new fit or global-optimum claim.

S12 Claim-limit map

In PC-9 RRA (three fresh paired rounds; median paired ratio), complete-workflow comparisons cover 18,187 common targeting genes, retained Control and all-zero rows reflect output conventions rather than biological discoveries, and top-rank overlap does not establish identical FDR calibration. In PC-9 MLE (six observations per engine; ratio of medians), very high treatment beta agreement does not imply exact floating-point identity. In HCC827 MLE (six observations per engine; ratio of medians), strong genome-wide rank and sign agreement coexists with selected large coefficient and Wald-inference differences whose underlying mechanism in Legacy MAGeCK remains unresolved, and numerical convergence in MAGeCKrs does not establish biological truth.

Across all workflows, exact raw-count and adapter agreement establishes a shared input foundation rather than identical downstream gene-level estimates. Reported runtimes apply strictly to the measured campaign binaries and fixed datasets; PC-9 RRA and PC-9 MLE ingest different complete FASTQ inputs, and stage-level medians in PC-9 RRA are not additive. These results do not claim universal or 10-fold end-to-end speedups, a new biological hit set, energy or memory advantages, cross-platform support, or precompiled binary distribution readiness.

Display rounding conventions: Narrative text and the Abstract report median runtimes to one decimal place and speedup ratios to two decimal places, whereas Table 2and all supplementary tables preserve full recorded precision. All other scientific metrics—including p-values, FDRs, correlations, and coefficient gaps—retain their exact source strings, and ratios are never recomputed from rounded display values. No new benchmarks, model fits, statistical calculations, or figure modifications were performed during manuscript finalization.